AS MATHEMATICS - PURE 1 - DAY 9

Binomial expansion

Coefficients and selected terms

YOUR AIM TODAY

Expand positive integer powers and locate a required coefficient.

YOUR RULE

Read the notes and copy each worked step. Attempt every question before opening page 4. Mark in another colour and repair every error.

Student: Rohan

Start time: ____________ Finish time: ____________

Date: __________________

LEARN FROM ZERO

The ideas you need

The coefficients in $(a+b)^n$ come from combinations.

- General term: $C(n,r)a^{(n-r)}b^r$.
- r starts at 0, so the $(r+1)$th term uses r.
- Signs alternate for $(a-b)^n$.
- For a selected coefficient, track both number and power of x.

MEMORY RULE

Write $C(n,r)$, then the first term power falls while the second rises.

Expansion

WORKED STEPS

$(1+2x)^3=1+3(2x)+3(2x)^2+(2x)^3=1+6x+12x^2+8x^3$.

Selected term

WORKED STEPS

x^2 in $(2+x)^5$ uses $r=2$: $C(5,2)2^3x^2=80x^2$.

YOUR TURN

Independent practice

Show every step. Include units in Mechanics.

Question	Working and final answer
1. Expand $(1+x)^4$.	
2. Expand $(2-x)^3$.	
3. Find coefficient of x^2 in $(1+3x)^5$.	
4. Find coefficient of x^3 in $(2+x)^6$.	
5. Find first four terms of $(1-2x)^5$.	
6. Find constant term in $(x+2/x)^4$.	

STOP

Do not continue until all six questions have an attempt.

ANSWER SECTION

Mark, diagnose, correct

A wrong answer is useful only when you repair the first wrong step.

Q	Answer and essential working
1	1+4x+6x^2+4x^3+x^4.
2	8-12x+6x^2-x^3.
3	C(5,2)3^2=90.
4	C(6,3)2^3=160.
5	1-10x+40x^2-80x^3.
6	Power x^(4-r)x^(-r)=x^(4-2r); constant when r=2. C(4,2)2^2=24.

Q	First wrong step	Correct rule

SEND FOR MARKING

Send pages 3 and 4. Ask for a score, the first error in each answer, and one follow-up question per weak skill.